Paulsen's Lectures on Quantum Computing and Completely Positive Maps, Formalised in Isabelle/HOL

with an assessment of the remaining gap to verified HHL

Dustin Bryant

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Abstract

We report on a formalisation in Isabelle/HOL of the mathematical content of Vern Paulsen's 2011 lecture course *Quantum Computing and Completely Positive Maps* (34 lectures): finite-dimensional Hilbert space theory, the postulates of quantum mechanics for pure and mixed states, partial traces, completely positive maps with Choi's theorem and the Choi–Kraus representation, the extreme points of the sets of unital and trace-preserving completely positive maps (Choi's Theorem 3), the Hahn–Banach theorem on matrix spaces, and the separating-hyperplane theory of entanglement witnesses. The development comprises sixteen theories and some nine hundred checked statements on top of the Jordan-Normal-Form and Isabelle Marries Dirac libraries; it builds without `sorry` or axioms and passes an oracle audit. A lecture-by-lecture audit records, for each statement in the notes, its formal counterpart or the reason it is absent, and an Isabelle-generated document reproduces the notes with the formal statements in place of the informal proofs. We then ask what this body of verified mathematics contributes towards a machine-checked proof of the Harrow–Hassidim–Lloyd (HHL) algorithm in Isabelle: the spectral theorem, the measurement calculus, tensor products and partial traces are in place and in a representation compatible with the existing Isabelle quantum libraries, while Hamiltonian simulation, phase estimation, the controlled eigenvalue inversion and the error analysis remain to be done.

1 Introduction

Machine-checked quantum computing in Isabelle/HOL has so far concentrated on algorithms: Isabelle Marries Dirac [2, 1] formalises qubits, gates, the no-cloning theorem, teleportation and the Deutsch and Deutsch–Jozsa algorithms on concrete matrices; the quantum Hoare logic of Liu et al. [9, 10] verifies Grover's algorithm; Manrique [11] adds the quantum Fourier transform. The mathematics that a graduate course on the subject spends most of its time on—positive matrices, the density-matrix formulation, partial traces, completely positive maps and their Choi matrices, the convex geometry of channels and of separable states—is covered unevenly. Unruh's libraries [3, 15, 16] develop much of this for arbitrary Hilbert spaces, in a type-based representation of operators; Echenim's entry [5] formalises projective measurements.

This paper describes a different kind of project: the end-to-end formalisation of one set of lecture notes, Paulsen's *Lectures on Quantum Computing and Completely Positive Maps* (Houston, Fall 2011) [12], taken as the specification. The notes are an unedited transcription of 34 lectures in five parts (basic mathematics; introduction to quantum computing; the theory of completely positive maps; entanglement; error correction). Following them lecture by lecture forces one to formalise the whole supporting theory rather than the shortest path to a headline theorem, and it produces a natural, externally fixed measure of completeness: the audit of Section 4.

The project started as an autoformalisation experiment and was finished by the author with AI assistance; everything reported here is checked by Isabelle2025-2 with the strict build option `quick_and_dirty = false`, and the principal theorems have been checked to depend on no oracles. The sources, the audit, and the generated document are available in the project repository.[1]

Contributions. (1) A verified formalisation of Parts I–IV of the notes and of the elementary part of Part V: in particular Schur's theorem, the spectral theorem for Hermitian matrices, four characterisations of positivity, Cholesky's reduction, the measurement postulates and POVMs, the characterisation of density matrices by ensembles, partial traces and their compatibility with local measurements, Choi's theorem and the Choi–Kraus representation of channels, the adjoint duality between unital and trace-preserving maps, Choi's Theorem 3 on extreme points, the Hahn–Banach theorem for matrix spaces, and the entanglement-witness characterisation of separability. (2) A lecture-by-lecture audit that classifies every definition and result of the notes as formalised, formalised in a variant form, not formalised, or deliberately omitted. (3) A document, generated by Isabelle from the theories, that restates the notes with the formal statements replacing the informal proofs. (4) An assessment of how far this material goes towards a verified HHL algorithm.

2 The notes and the formalisation method

Representation. All objects of the notes are concrete: vectors in $\mathbb{C}^n$ and matrices in $M_{m,n}$. We use the Jordan-Normal-Form library [14]: vectors are complex Matrix.vec with a dimension hypothesis dim_vec v = n, matrices are complex Matrix.mat with carrier hypotheses A ∈ carrier_mat m n. Dimensions are hypotheses rather than types, which is how the notes treat them and which is the convention of Isabelle Marries Dirac, so that its gates, inner product, adjoint (dagger) and tensor products can be reused directly. Maps $\Phi : M_n \to M_d$ are HOL functions complex Matrix.mat ⇒ complex Matrix.mat constrained only on the carrier; linearity, positivity and complete positivity are predicates on such functions.

Two places in the notes genuinely need topology (compactness and the separation of a point from a closed convex cone, Days 22 and 27–28). For these we introduce a typed wrapper ('a,'m,'n) lmat of matrices of fixed dimension, instantiate the HOL-Analysis type classes up to euclidean_space and heine_borel, and transport the relevant statements across the wrapper. The carrier-predicate world and the typed world coexist; the former carries almost all of the development, the latter is used only where HOL-Analysis is applied.

Fidelity. Statements are formalised as close to the notes as the representation permits. The audit distinguishes faithful statements from *variants*; the main variants are: the predicate psd carries Hermiticity as an explicit conjunct (equivalent by the notes' own Theorem that $\langle x|Hx\rangle \in \mathbb{R}$ for all x iff $H = H^*$); Cholesky's reduction adds the hypothesis $p_{11} > 0$ that the notes leave implicit; linear maps are given with explicit additivity and homogeneity hypotheses on the carrier; and Choi's Theorem 3 is proved in its trace-preserving form with a fixed Kraus list of linearly independent operators, see Section 3. Where the notes themselves are inconsistent we formalise the correct statement and say so in the commentary (the notes claim $H^2 = X$ on Day 10 and correct it to $H^2 = I$ on Day 11; they use two sign conventions for the Pauli matrix Y).

Verification discipline. The project builds as three Isabelle sessions with `quick_and_dirty = false`, so no `sorry` can survive. We also ran an oracle audit (Thm_Deps.all_oracles over the principal theorems). This turned up a methodological point worth recording: under

[1] `https://github.com/brydustin/Quantum_Computing`

Theory	Lectures	lines	resu
Basic_Math	1–2: inner products, matrix units, trace, projections, Parseval	1 672	
Quantum_Computing_3_10	3–10: unitaries, Schur, spectral theorem, positivity, tensors, postulates	7 945	1
Quantum_Computing11	11–22, 29–32: channels, density matrices, Choi, convexity, codes	11 217	3
Day23_ChoiExtremeUCP	23: Choi's Theorem 3, CP Radon–Nikodym	7 001	1
Matrix_Typed	25: Hahn–Banach on matrix spaces	1 346	
Essential_Functional_Analysis	Hilbert–Schmidt analysis, Carathéodory	2 536	
Lmat_Euclidean	typed matrices as a Euclidean space	780	
Separable_Cone	27–28: separable cone, witnesses	3 107	
Day24_Separability	26–28 restated lecture by lecture	210	
QC_Prelims	shared matrix and vector lemmas of the gap-closure pass	315	
Day12_Counting_Probability	12: $H^{\otimes N}\|0\cdots0\rangle$, the counting probability	267	
Day18_Positive_Maps	17–18: density-preserving iff positive and TP; the transpose	257	
Day22_Kraus_Independence	22: $\mathrm{cr}(\Phi) \leq n$ for extreme CPTP maps	181	
Day28_Horodecki_Peres	28: the two-qubit transpose criterion	251	
Day30_BitFlip_Code	30: syndromes, recovery map, the recovery theorem	441	
Day32_Pauli_Expansion	32: $\sum \lvert a_k\rvert^2 = 1$ for Pauli expansions of unitaries	59	

Table 1: The sixteen theories after the commentary and gap-closure passes (results = lemmas, theorems, corollaries, propositions; 934 in all). The lower block closes the medium-sized gaps of the first audit.

`quick_and_dirty = true`, Isabelle's Goal.prove_sorry discharges package-internal goals of fun and record definitions with Skip_Proof.cheat_tac, so every theorem downstream of such a definition reports the oracle Pure.skip_proof even in a development that contains no `sorry` at all. Oracle audits are only meaningful on builds with the strict option.

3 What is formalised

Table 1 lists the theories. We describe the content by lecture group; the names in brackets are the Isabelle theorems.

Lectures 1–10: linear algebra and the postulates. Besides the inner-product algebra, matrix units and the trace (Tr_cyclic, hs_inner_prod_as_trace), the first part proves the projection formula $P = \sum |v_i\rangle\langle v_i|$ and Parseval (parseval_norm), the characterisations of unitaries, Householder unitaries and Schur's lemma (schur_lemma), Schur's triangularisation (schur_theorem) and $\mathrm{Tr}\, A = \sum \lambda_i$, the spectral theorem for Hermitian matrices (hermitian_spectral_decomp: $U^\dagger H U$ is a real diagonal matrix), and the positivity theory: $P \geq 0$ iff Hermitian with non-negative eigenvalues (psd_iff_nonneg_eigenvalues), iff $P = \sum |v_i\rangle\langle v_i|$ (psd_kraus_decomp, outer_sum_psd), positive definiteness iff the v_i span (pd_iff_spanning), the positive square root, and Cholesky's reduction (cholesky_reduction). Tensor products are Kronecker products with bilinearity, the multiplicativity of inner products and the orthonormal product basis (tensor_onb). The postulates are definitions: quantum_state, evolve, measurement_system, meas_prob, with the probabilities summing to one, POVMs with their square-root measurement systems (povm_from_sqrt), perfect distinguishability of orthogonal states, and local measurements in composite systems (composite_measurement_prob).

Lectures 11–22: channels and Choi's theorem. The no-cloning theorem (no_cloning) and the existence of oracles U_f (oracle_exists) open the second part. Ensembles and their density matrices are defined, and density matrices are characterised as the positive matrices of trace one (density_matrix_iff_ensemble, via the spectral ensemble); the ensembles of a

pure state consist of multiples of the state (pure_ensemble_parallel). Partial traces are defined by the block formulas of the notes; they are trace preserving and positivity preserving (partial_trace_B_is_density) and compatible with local measurements: $\mathrm{Tr}((M^\dagger M \otimes I)\rho) = \mathrm{Tr}(M^\dagger M \,\mathrm{tr}_B \rho)$ (partial_trace_measurement). The coordinate form of $L(\mathcal{H}_A) \otimes L(\mathcal{H}_B) \cong L(\mathcal{H}_A \otimes \mathcal{H}_B)$ is an explicit linear isomorphism. Measurement maps $\rho \mapsto \sum M_\alpha \rho M_\alpha^\dagger$ preserve density matrices; Kraus maps are completely positive (kraus_is_cp), with complete positivity defined through the block amplification $\Phi^{(p)}$. Choi's theorem is choi_theorem: a linear Φ on the carrier is completely positive iff its Choi matrix $(\Phi(E_{ij}))_{ij}$ is positive, and kraus_rep_of_psd_choi extracts a Kraus representation with nd operators from a positive Choi matrix, which gives the Choi–Kraus characterisation of channels (choi_kraus_cptp) and the Kraus-rank bound (kraus_rank_bound). The Hilbert–Schmidt adjoint of a Kraus map is the Kraus map of the adjoint operators (kraus_adjoint_correct), and Φ is unital iff Φ^* is trace preserving (cp_unital_iff_adjoint_tp); positive functionals are completely positive (positive_scalar_is_cp); the sets UCP and CPTP are convex.

Lecture 23: Choi's Theorem 3. Choi's characterisation of the extreme points of UCP [4] is formalised in its trace-preserving form, for a channel given by a Kraus list $V_1, \ldots, V_r$ of linearly independent operators (choi_theorem_3_cptp_iff): Φ is extreme if and only if $\{V_j^\dagger V_i\}$ is linearly independent over $\mathbb{C}$. (The perturbation argument of the notes produces a real-coefficient relation; a complex relation is first turned into a Hermitian one, using that the family $\{V_j^\dagger V_i\}$ is closed under adjoints, so both directions use the same notion of independence.) Two aspects deserve comment. First, because HOL functions are only constrained on the carrier, the naive definition of an extreme point ("Φ is not the midpoint of two *different* CPTP maps") is vacuous: any CPTP map agrees on the carrier with maps that differ from it elsewhere. Extremality is therefore defined through disagreement on the carrier (extreme_CPTP_carrier). Second, the notes' proof uses the unitary freedom of Kraus representations (their "Choi 2"), which is not formalised. The forward direction instead goes through a CP Radon–Nikodym theorem (cp_radon_nikodym_linear): if Ψ and $\Phi - \Psi$ are completely positive and $\Phi(X) = \sum_i V_i X V_i^\dagger$, then $\Psi(X) = \sum_{ij} T_{ij} V_i X V_j^\dagger$ for a positive matrix T. Its proof reduces to a statement about positive matrices dominated by an outer sum, $0 \le B \le VV^\dagger$ implies $B = VTV^\dagger$ with $T \ge 0$, which we prove with the Moore–Penrose pseudoinverse of $VV^\dagger$ built from the spectral theorem: the identities $AA^+A = A$ and, for $0 \le B \le A$, $AA^+B = B$ follow from the kernel inclusion $\ker A \subseteq \ker B$, and $T = (A^+V)^\dagger B (A^+V)$.

Lectures 25–28: Hahn–Banach and entanglement witnesses. The notes state the Hahn–Banach theorem for bounded functionals on normed spaces and use it in Arveson's extension theorem. We prove it for matrix spaces with the Hilbert–Schmidt norm (matrix_hahn_banach_lmat): HOL's Hahn–Banach theorem gives a norm-preserving real-linear extension of the real part, and the Bohnenblust–Sobczyk formula $f(x) = F(x) - iF(ix)$ recovers a complex-linear extension with the same bound. Arveson's correspondence and extension theorem are not formalised. The entanglement theory of Days 27–28 is: the separable matrices form a closed convex cone (separable_cone_lmat_closed, using the closedness of the positive matrices, the compactness of the product density matrices and a Carathéodory bound on the number of generators), so a non-separable ρ is separated from the cone by a real functional, which becomes a Hermitian witness W with $\Re\,\mathrm{Tr}(W\rho) < 0 \le \Re\,\mathrm{Tr}(W\sigma)$ for all separable σ (entanglement_witness_hermitian'); this yields the Horodecki-type characterisation [8]: a Hermitian ρ is separable iff every such witness is non-negative on it (separable_iff_witness), with a strict version for density matrices. The notes phrase the criterion through positive maps, which needs the Arveson correspondence; the witness form is the equivalent statement that does not.

Lectures 29–32: error correction. Hamming weight and distance, the parity check and repetition codes, the Pauli matrices and their orthogonality, the bit-flip, phase-flip and bit-phase-flip channels (as UCP and CPTP maps), the pure-state fidelity, and the code words of the three-qubit codes are formalised. The recovery theorems for the three-qubit code and the Shor code, and the Knill–Laflamme conditions of the last lectures, are not.

4 The audit

The file `NOTES_AUDIT.md` in the repository lists, for each of the 34 lectures, every definition, theorem, proposition, corollary, example and remark of the notes with its formal counterpart. Each item is classified: ✓ faithful, ∼ formalised in a variant form, × not formalised, or − deliberately not formalised (overviews, history, numerical examples, classical asides such as stochastic matrices or cyclic codes). Of the 170 items, 11 are deliberate omissions; of the remaining 159 mathematical items, 101 are faithful, 20 are variants and 38 are gaps.[2] The audit drove a final round of proofs: small statements omitted along the way (the converses isometry ⇒ unitary, positive definiteness iff positive eigenvalues, the expected value $\langle\psi|H\psi\rangle$ of an observable, the phase invariance of $|\psi\rangle\langle\psi|$, that $\rho_1 \otimes \rho_2$ is a density matrix, the unambiguous discrimination of linearly independent states, and the complex-coefficient form of Choi's Theorem 3) were proved during the close-out and are counted as faithful above. A second pass closed the medium-sized gaps that the notes prove in a page, each in its own theory (the lower block of Table 1): a linear map sends density matrices to density matrices iff it is positive and trace preserving (density_preserving_iff_positive_tp); the transpose on M_2 is positive but not 2-positive, with the notes' witness (E_{ij}); the Hadamard power $H^{\otimes N}|0\cdots0\rangle = 2^{-N/2}\sum_J |J\rangle$ and the counting probability $p_0 = (M/2^N)^2$ of Day 12, for any unitary acting as the oracle on the basis; the Pauli expansion identity $\sum |a_k|^2 = 1$ for unitaries; the bound $\mathrm{cr}(\Phi) \le n$ for extreme CPTP maps (linear independence bounds the length, through the vector-space library of the Jordan-Normal-Form entry); the two-qubit Horodecki–Peres criterion, $(\mathrm{id}\otimes t)(|v\rangle\langle v|) \ge 0$ iff v is a product vector, with the witness value $-2|v_0v_3 - v_1v_2|^2$; and the three-qubit bit-flip code, whose syndrome projections, CPTP recovery map and recovery theorem $R(|\psi_i\rangle\langle\psi_i|) = |\psi\rangle\langle\psi|$ for at most one flip are proved on explicit 8×8 matrices. Substantial developments remain: the Choi rank theorem $\mathrm{cr}(\Phi) = \operatorname{rank} P_\Phi$ and the unitary freedom of Kraus representations; operator systems with Arveson's correspondence and extension theorem; the Stinespring environment model of noise; the Shor code and the Knill–Laflamme conditions.

Beyond the omissions, the audit documents where the formalisation strengthens the notes (strict witnesses for density matrices; Cholesky with the missing hypothesis; the vacuity of the naive extreme-point definition) and where it changes the route of a proof (Choi's Theorem 3 through CP Radon–Nikodym).

5 The generated document

Isabelle's document preparation turns the theories into a PDF in which proofs are folded and the commentary appears between the formal statements. During this project we rewrote the top-level comments of all theories as document markup: each lecture is a section, and before each group of formal statements a text block restates the notes' definitions and results in their own words and LaTeX, says which Isabelle statements correspond to them and how they differ, and names the audit items for what is missing. The result is a 2011 lecture course in which every displayed theorem has been checked, organised by theory rather than strictly by date because later lectures reuse earlier infrastructure; a reading guide at the front gives the map from lectures to theories. The document is produced by a fourth session, Vern_Paulsen_QC_Notes,

[2] Counts as of the second close-out of October 2026; the audit file is the authoritative record.

through Isabelle's `document_theories` mechanism, so it is regenerated and re-checked by every build.

6 Relation to the Archive of Formal Proofs

Isabelle Marries Dirac [1] and the quantum Fourier transform entry [11] use the same matrix representation as this project (complex Matrix.mat with carriers), and we import both; our tensor-product lemmas (mixed product, adjoint, unitarity, product bases), measurement calculus and density-matrix theory extend them in the same style. The quantum Hoare logic entry [10] contains a positivity predicate, the Loewner order and a decomposition of positive matrices, overlapping with our Days 4 and 18, but no Choi theory. Unruh's *Complex Bounded Operators*, *Tensor Products in Hilbert Spaces* and *Kraus Maps* [3, 15, 16] are far more general—arbitrary Hilbert spaces, trace-class and Hilbert–Schmidt operators, the spectral theorem for compact operators, Kraus families for possibly infinite-dimensional channels—but live in a type-based representation of operators ('a $\Rightarrow_{CL}$ 'b); they contain neither Choi's theorem nor the extreme-point and separability results, and connecting them to the matrix world would require a transfer layer of the kind our lmat wrapper provides in miniature. Echenim's projective measurements [5] are formulated on complex^'n vectors. To our knowledge the results of Days 19–28 (Choi's theorem in its matrix form with explicit Kraus extraction, Choi's Theorem 3, the Hahn–Banach theorem on matrix spaces, the closedness of the separable cone and the witness characterisation) have no counterpart in the Archive.

7 Towards a verified HHL algorithm

The HHL algorithm [7] prepares, for a Hermitian A with eigen-decomposition $A = \sum_j \lambda_j |u_j\rangle\langle u_j|$ and a unit vector $|b\rangle = \sum_j \beta_j |u_j\rangle$, a state proportional to $A^{-1}|b\rangle$: phase estimation with the unitaries e^{iAt} writes the eigenvalues into a register, a controlled rotation of an ancilla produces the amplitudes C/λ_j, uncomputation and a measurement of the ancilla postselect the branch $\sum_j (\beta_j/\lambda_j)|u_j\rangle$, up to the approximation error of phase estimation and with a success probability governed by the condition number. Machine checked proofs exist in Coq, in the quantum Hoare logic of Zhou, Yu and Ying [18, 17, 6]; none exists in Isabelle/HOL, whose verified algorithms are Deutsch–Jozsa, Grover and the QFT (a recent HOL Light development treats Grover [13]).

What a HHL proof needs, and what this project supplies in the Isabelle Marries Dirac representation:

- *The spectral theorem.* The whole analysis is carried out in the eigenbasis of A. hermitian_spectral_decomp provides the unitary U and the real list D of eigenvalues with $U^\dagger A U = \mathrm{diag}(D)$, together with the facts that the columns of U are an orthonormal eigenbasis and that $\beta_j = \langle u_j | b\rangle$ (Parseval). It also gives the functional calculus HHL needs: $f(A) = U\,\mathrm{diag}(f(D))\,U^\dagger$ is how psd_sqrt and the pseudoinverse pinv_list are built, and A^{-1} on the range of A is literally the pseudoinverse whose identities $AA^+A = A$ we prove.

- *Unitarity and Hamiltonian simulation.* $e^{iAt} = U\,\mathrm{diag}(e^{i\lambda_j t})\,U^\dagger$ can be defined through the same decomposition; its unitarity is the lemma that a conjugate of a diagonal unitary by a unitary is unitary, which is a three-line consequence of unitary_mult and the characterisation of unitaries. What is missing is the connection to an independently defined matrix exponential and the Trotter-type approximation results; for the correctness of HHL as an idealised circuit the spectral definition suffices.

- *Registers, tensor products, controlled operations.* Multi-register states are Kronecker products; the lemmas tensor_mat_mult, tensor_mat_adjoint, tensor_mat_unitary and tensor_onb are the algebra of composite registers. Controlled unitaries are block matrices (four_block_mat, with the block-action lemma block_mat_apply). The QFT itself is available from the Archive in the same representation [11]; quantum phase estimation is not yet formalised in Isabelle and is the largest missing circuit-level component.

- *Measurement and postselection.* The ancilla measurement is a two-outcome measurement system; meas_prob gives the success probability and post_meas_state the normalised output, and composite_measurement_prob together with partial_trace_measurement express that measuring the ancilla alone is a local measurement whose statistics are those of the reduced state. The POVM formalism and the density-matrix postulates allow the same analysis for mixed inputs and for a noisy implementation, where the CP-map theory (Kraus representation, Choi matrix, convexity of CPTP) is the natural language.

- *Error analysis.* The phase-estimation error, the dependence of the success probability on the condition number κ, and the final fidelity bound are analytic estimates that the notes do not touch; the Hilbert–Schmidt norm theory of Essential_Functional_Analysis and the fidelity of Day 31 are starting points, but this part of a HHL proof would have to be built from scratch.

In short, the linear-algebraic and measurement-theoretic layer of a HHL proof—the part that a textbook treats in one line, "work in the eigenbasis of A"—is now available and verified in the representation that the existing Isabelle quantum libraries use. The remaining gap is concentrated in three places: a formalisation of quantum phase estimation on top of the QFT entry, the controlled eigenvalue-inversion step with its amplitude bookkeeping, and the quantitative error analysis. Each is a project of a size comparable to one of the lecture groups above.

8 Conclusion

Taking a complete set of lecture notes as the specification turned out to be a productive way to build a library: it fixes the scope, orders the work, and makes completeness measurable. The cost is visible in the audit: a course mentions many results it does not prove, and a formalisation faithful to the notes must decide, item by item, what to prove, what to restate, and what to leave. The commentary pass that produced the generated document is the record of those decisions; we recommend it as a deliverable for any formalisation of a text, because it is the only place where the informal and the formal statements are read side by side.

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